Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	The Cooperative Multi-Robot Gantry (CMRG): An AI-Enabled Platform for Large-Scale Manufacturing, Assembly, and Inspection of Modular Building Structures and Components			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical 、 Materials		Mechanical 、 Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	*Robotics/ Automation Machinery*	Measurement/Sensor Technologies for Automation
Concept of Proposed Technology	A defining feature of the proposed technology is its ability to enable true multi-robot construction processes, in which two robotic arms operate cooperatively on tasks that cannot be accomplished by a single robot alone. This includes coordinated manipulation of large or heavy structural modules, cooperative alignment and positioning, and synchronized execution of assembly, connection, and inspection operations. By operating at realistic construction scales, the system supports the fabrication, assembly, and disassembly of full-size modular structural components within a prefabrication and modular construction context, rather than reduced laboratory prototypes.			

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The gantry-based architecture is intentionally designed as a flexible, reconfigurable research and deployment platform, allowing the exploration of diverse construction processes, material systems, and structural typologies. Interchangeable tooling, adaptable sensing, and software-defined workflows enable the same system to support multiple materials (e.g., steel, timber, hybrid systems), different connection strategies, and evolving construction methods. This flexibility ensures that the platform can serve both as a validation environment for near-term construction automation workflows and as a testbed for emerging multi-robot fabrication and assembly paradigms.

Key Features:

- Cooperative multi-robot construction at realistic structural scales
  Enables two robotic arms to operate collaboratively on tasks that cannot be performed by a single robot alone, including coordinated handling, alignment, and assembly of large and heavy structural modules. The system is designed to execute construction workflows at full, construction-relevant scales within a modular prefabrication context.
- Flexible gantry-based platform for diverse materials and structural systems
  A large-workspace gantry architecture with reconfigurable tooling and software-defined workflows allows exploration of multiple construction processes, material systems (e.g., steel, timber, hybrid structures), and structural typologies, supporting both near-term deployment and exploratory research.
- Tool and hardware development for large-scale assembly and inspection
  Design and fabrication of robust, interchangeable end-effectors for gripping, alignment, connection, and inspection operations enable reliable automation of large structural components and repeatable quality assurance in practical construction environments.
- Vision–language–based perception and constraint-aware reasoning
  Vision–language models interpret scene context, task intent, and construction-relevant constraints—such as geometry, access limitations, safety zones, tolerances, and component states—to support high-level reasoning and constraint-aware decision-making during planning and execution.
- Simulation-based reinforcement learning for construction planning and optimization
  High-fidelity simulation combined with reinforcement learning is used to evaluate construction designs, optimize assembly and disassembly sequences, and compute efficient construction methods and schedules under real-world constraints prior to deployment on physical hardware.
- Closed-loop sensing, verification, and adaptive execution
  Integrated force–torque sensing, vision, and inspection systems provide continuous feedback during fabrication and assembly, enabling adaptive control, tolerance compensation, and automated verification of construction quality and task completion.

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Available Equipment for Construction-Scale Automation and Inspection:

Figure 1. Large-Workspace Gantry Testbed and Example Task Scenario (Dual ABB IRB 6620 Arms)

- Figure 1 illustrates the large-scale cooperative robotic gantry platform, equipped with two ABB IRB 6620 industrial robotic arms. The system provides approximately 17.0 m of longitudinal travel, a 9.5–10.4 m gantry span, and 4.2–4.8 m of vertical clearance, enabling full-scale validation of autonomous assembly, disassembly, and construction-scale material handling and manipulation workflows.
- The platform is instrumented with integrated heavy-payload six-degree-of-freedom force–torque sensors mounted on the robot side of the tool quick-changer. These sensors provide real-time force–torque measurements during fabrication and assembly processes and support adaptive planning, compliance-based control, and process monitoring.
- Automatic tool quick-changers with full electrical, pneumatic, and data pass-throughs enable rapid tool exchange, allowing the system to transition efficiently between fabrication, handling, and assembly tasks.
- The platform includes customized end-effectors, including grippers, nailing tools, and other application-specific tooling designed for large-scale structural assembly.
- Finally, distance and proximity sensors mounted on end-effectors provide feedback on existing geometry and site conditions, enabling closed-loop control, tolerance compensation, and adaptive execution of construction processes.

Rationale for Joint R&D Project

The proposed project relies on **close collaboration between an academic research institute and industrial partners** to develop and validate **real-world, construction-scale test cases** for AI-driven robotic manufacturing, assembly, and inspection. Rather than targeting immediate field deployment, the focus is on using a large-workspace gantry-based robotic system as a **unique experimental testbed** for off-site modular construction and inspection workflows. This joint R&D framework enables advanced autonomy algorithms, novel hardware, and new construction processes to be tested and refined under realistic geometric, material, and operational constraints that cannot be replicated in small-scale laboratory environments.

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Rationale for Collaboration

- **Georgia Tech (PI Bruun)** provides construction and civil engineering expertise, construction automation experience, and access to the Cooperative Multi-Robot Gantry (CMRG) system. This includes knowledge of constructability constraints, structural handling and assembly, and inspection methodologies for construction materials and welded joints. PI Bruun also brings prior experience in **human-in-the-loop robotic construction and augmented-reality (AR)–based interfaces** for shared planning, guidance, and supervision. The proposed collaboration enables these approaches to be **scaled to new scenarios**, including larger structural modules, cooperative multi-robot tasks, and more complex assembly and inspection workflows than have previously been explored.
- **The Korean SME and startup partners** contribute expertise in autonomy software development, including vision–language-based perception, constraint-aware reasoning, decision-making, and simulation-based learning and optimization. These capabilities support both fully autonomous operation and **shared-autonomy modes**, in which human input—delivered through interfaces such as AR—can be incorporated for supervision, exception handling, and task-level guidance.
- **The CMRG enables experimentation at realistic construction scales**, allowing industrial partners to test processes, software, sensing approaches, and hardware concepts on full-size modular structural elements rather than reduced-scale prototypes. This capability is particularly valuable for startups and SMEs seeking to validate new robotic construction processes, hardware designs, or autonomy concepts before commercialization.
- **The platform's flexibility supports exploration of diverse materials, processes, and interaction modes**, including steel, timber, concrete, and hybrid systems, as well as autonomous, human-supervised, and mixed-initiative construction workflows. Automatic tool changing, modular interfaces, and software-defined control enable rapid reconfiguration across experiments.

Expected Benefits of Joint R&D

- **Validated construction-scale test cases** for AI-driven robotic manufacturing, assembly, disassembly, inspection, and human-robot collaborative workflows in off-site modular construction contexts.
- **De-risking of new robotic construction processes** through large-scale experimental validation that bridges simulation, small-scale testing, and future industrial adoption.
- **A shared experimental platform for collaboration with startups and Korean companies**, enabling co-development of autonomy software, sensing systems, AR-enabled human-in-the-loop interfaces, and construction-ready hardware at realistic scales.
- **New insights into scalable human–robot collaboration**, including the role of augmented reality and shared autonomy in improving safety, adaptability, and performance in large-scale construction robotics.

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PI Bruun's Past Research Projects:

Figure 2. Prior Cooperative Robotic (top) and Augmented-Reality (bottom) Construction Projects

Figure 2 presents representative case studies of cooperative robotic construction in timber and masonry, highlighting multi-robot coordination for construction-scale assembly, disassembly, and scaffold-free fabrication. In these workflows, one robot executes primary placement or building operations while a second robot provides temporary structural support and performs complementary tasks, enabling stable, sequential construction of complex structural systems. The lower examples illustrate augmented-reality–assisted construction for geometrically complex forms, including a double-curved masonry brick vault, where AR provides real-time spatial guidance and feedback to support accurate placement and execution of non-standard geometry.

Key Words

1. AI-Enabled Modular and Prefabricated Construction
2. Cooperative Multi-Robot Gantry Systems
3. Construction-Scale Robotic Assembly and Disassembly
4. Robotic Inspection and QA/QC of Modular Structural Components
5. Vision–Language–Based Constraint-Aware Planning

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop an AI-driven gantry-based construction automation system that can reliably execute end-to-end autonomous construction workflows in realistic construction environments by integrating large-workspace manipulation, construction-specific tooling, perception-based constraint reasoning, and automated post-process inspection. The specific R&D objectives are: • To develop construction-specific tools and hardware that enable reliable large-scale assembly, disassembly, and post-process inspection in practical environments. • To develop vision-language-enabled perception and constraint reasoning that supports decision-making, planning, and safe execution under real construction constraints. • To develop a high-fidelity simulation and reinforcement-learning framework to optimize constructability and efficient construction sequences before deployment on hardware. • To demonstrate and validate an integrated end-to-end autonomous “plan, build, inspect” workflow through laboratory and pilot-scale testing under realistic conditions.			
R&D Target Specifications	**Closed-Loop Tool and Hardware Manipulation Accuracy on Representative Structures (< 5 mm):** Demonstrate reliable operation of construction-specific end-effectors by achieving gripping, alignment, and connection accuracy within 5 mm (or tighter, as required by the process) on representative structural assemblies. Accuracy shall be achieved and maintained through closed-loop feedback during execution, incorporating real-time sensing (e.g., force–torque, vision, proximity) to detect and correct deviations as tolerances manifest during construction, rather than relying solely on open-loop positioning. **Robustness of Multi-Robot Task-and-Motion Planning (> 90%):** Achieve a planning success rate exceeding 90% across diverse simulated construction scenarios by autonomously generating coordinated, collision-free task and motion plans for cooperative multi-robot processes. Planning shall explicitly account for robot–robot interactions, shared workspaces, handoff operations, tool changes, and sequencing constraints, as well as geometric, kinematic, and operational limitations. Robustness is demonstrated by successful plan generation and execution under varying layouts, component geometries, and constraint sets. **End-to-End Autonomous Construction of Complex Geometries at Construction Scale:** Demonstrate autonomous planning and execution of complete construction workflows for structurally meaningful, nontrivial geometries at construction-relevant scales. The system shall fabricate and/or assemble modular structural components with characteristic dimensions of 3–8 m and individual component masses of up to 1–3 metric tons, depending on material and process. Demonstrations may include wire-based additive manufacturing, concrete 3D printing, and timber milling and member assembly. Performance shall include coordinated multi-robot manipulation and execution with a global geometric deviation of ≤ 10 mm relative to the design model, validated on full-scale modules under realistic workspace, handling, and tolerance constraints. This R&D project will be conducted over **three years**, following a **progressive development and validation strategy** that advances from collaborative ideation and simulation-based workflow development to laboratory-scale physical experimentation, and finally to **full-scale pilot demonstrations at realistic construction scales**.			

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R&D Plan	**Year 1: Collaborative Ideation and Foundations — Leveraging Cooperative, Large-Scale Robotics** Year 1 focuses on **close collaboration with SME and startup partners** to jointly develop and refine construction automation concepts that **explicitly leverage the unique capabilities of the Cooperative Multi-Robot Gantry (CMRG)**—namely, cooperative multi-robot operation, large workspace coverage, and heavy-payload manipulation that are not achievable in conventional industrial robot cells. • Conduct structured **co-design and ideation workshops** with SME and startup partners to identify construction workflows, materials, and processes that benefit from **multi-robot cooperation**, long-reach manipulation, and high payload capacity. • Prioritize use cases that require **coordinated handling of large or heavy structural components**, synchronized assembly operations, or construction-scale fabrication processes that cannot be meaningfully evaluated at reduced scale. • Define representative construction materials, component types, connection operations, and practical constraints, including geometry, tolerances, safety rules, workspace limits, and access conditions, grounded in the selected large-scale use cases. • Translate prioritized concepts into **end-to-end construction automation workflows**, integrating perception, constraint reasoning, task planning, motion planning, and cooperative execution. • Integrate new sensing concepts, hardware abstractions, and computational processes required to support cooperative, large-scale workflows. • Design and fabricate initial construction-specific tools and end-effectors for gripping, alignment, and basic connection operations to support early hardware-in-the-loop testing. • Build a high-fidelity simulation environment for gantry-based **cooperative multi-robot construction** under the defined constraints. • Develop vision–language–enabled constraint reasoning and reinforcement-learning-based task-and-motion planning, and benchmark performance across diverse simulated scenarios that reflect the scale and cooperation demands of the CMRG. **Year 2: Laboratory Integration — Small-Scale Execution and Human-in-the-Loop Feedback** Year 2 transitions from simulation to **laboratory-scale physical experimentation**, focusing on executing the co-designed workflows on smaller structural components and subassemblies while developing **closed-loop and human-in-the-loop feedback mechanisms**. • Integrate and refine interchangeable end-effectors on the gantry platform and robotic arms, and validate tool performance on representative structural components and subassemblies. • Execute construction workflows on small-scale structures, verifying constraint recognition, safe execution, and coordination between perception, planning, and control. • Develop and validate **closed-loop error detection and correction pipelines**, combining onboard sensing with **human-in-the-loop input**, including augmented-reality-based guidance for alignment correction, exception handling, and task verification. • Integrate post-process inspection workflows for construction materials, enabling repeatable quality evaluation and reporting. • Quantitatively evaluate manipulation accuracy, execution robustness, and the effectiveness of human-assisted versus fully autonomous correction strategies through repeated laboratory experiments. **Year 3: Full-Scale Pilot Demonstrations — Construction-Scale Validation** Year 3 focuses on **full-scale pilot demonstrations**, validating the complete system on construction-relevant geometries, component sizes, and handling conditions. • Execute cooperative multi-robot construction workflows on **full-scale modular structural components**, including fabrication, assembly, disassembly, and inspection.

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	• Demonstrate end-to-end **"plan–build–verify" workflows** that integrate autonomous operation with **human-in-the-loop supervision and augmented-reality-enabled feedback**. • Validate performance on complex geometries and heavy components under realistic workspace, tolerance, and safety constraints. • Assess system scalability, robustness, repeatability, and the role of human input across multiple full-scale pilot demonstrations. • Finalize performance benchmarks, experimental protocols, and best-practice guidelines to support continued collaboration with industry partners and future scaling.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Expected R&R of the Research Institute (Georgia Institute of Technology)** The research institute will be responsible for the **physical implementation, integration, and experimental validation** of the proposed AI-driven gantry-based construction automation system. The institute will lead construction-oriented system development, large-scale experimentation, and performance evaluation, using the existing Cooperative Multi-Robot Gantry (CMRG) platform. Key responsibilities include: • **Physical system integration and experimental operation** Lead system setup, calibration, and operation of the gantry platform and robotic arms, including safety protocols, multi-robot coordination, data acquisition, and quantitative benchmarking under realistic construction conditions. • **Development and integration of construction-specific tooling and sensing** Design, fabricate, and integrate interchangeable end-effectors and sensing modules for gripping, alignment, assembly, fabrication, and inspection processes across multiple material systems. Tooling developed by the SME partner will be physically integrated, tested, and validated within the CMRG workflow. • **Closed-loop construction execution and feedback pipelines** Develop and validate closed-loop construction workflows that incorporate real-time sensing, error detection, tolerance compensation, and corrective actions during execution. This includes integrating robotic control, multi-robot coordination, and adaptive execution informed by live data. • **Human-in-the-loop and augmented-reality-enabled processes** Design and evaluate human-in-the-loop construction workflows, including augmented-reality-based supervision, guidance, and exception handling, and study their role in improving robustness, safety, and performance at large construction scales. • **Definition of construction-relevant constraints, test scenarios, and metrics** Define representative construction constraints, experimental scenarios, and performance metrics, and provide construction and civil engineering expertise to guide integrated development and evaluation with the industrial partner.
Expected R&R of Korean SME Partner	The project follows a **tight hardware–software co-development model**, in which the Korean SME leads autonomy, planning, and application definition, while the research institute leads physical implementation, execution, sensing, and experimental validation. Continuous iteration between simulation and full-scale experimentation ensures that algorithmic developments are grounded in real construction constraints and that physical insights directly inform planning and design. **Expected R&R of the Korean SME Partner:** The Korean SME partner will be responsible for algorithmic autonomy development, planning software, and large-scale application definition, with a focus on enabling full or shared autonomy under realistic construction constraints. Key responsibilities include:

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	• **Development of autonomy and planning algorithms:** Develop autonomy software modules for construction automation, including vision–language–enabled perception, constraint-aware reasoning, decision-making, and task-level logic for complex construction workflows. • **Simulation and learning-based planning frameworks:** Develop high-fidelity simulation environments and reinforcement-learning-based task-and-motion planning methods to optimize construction sequencing, multi-robot coordination, and execution strategies under diverse constraints. • **Software integration and system-level interfaces:** Integrate autonomy software with the robotic system workflow, including interfaces for planning-to-execution handoff, monitoring, logging, and performance evaluation, in coordination with the research institute. • **Definition of large-scale construction applications and processes:** Propose and develop construction applications that benefit from the cooperative, large-workspace, and heavy-payload capabilities of the CMRG, including novel fabrication, assembly, or inspection processes to be tested at realistic scales. • **Design of custom tooling and hardware concepts:** Design custom tools, sensors, or hardware concepts required for their proposed applications. These components will be transferred to the research institute for physical integration, testing, and validation within the gantry-based construction workflows.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed joint R&D project will deliver an AI-driven, gantry-based construction automation prototype that enables end-to-end “plan–build–verify” workflows for large-scale modular construction and inspection. Centered on the Cooperative Multi-Robot Gantry (CMRG) platform, the project advances construction robotics by enabling cooperative multi-robot operation, heavy-payload manipulation, and construction-scale execution that cannot be achieved with conventional industrial robot cells. The system supports autonomous and shared-autonomy workflows for fabrication, assembly, disassembly, and inspection of modular components across multiple material systems, including steel, concrete, timber, and hybrid assemblies. Technically, the project integrates construction-specific interchangeable tooling, closed-loop sensing and feedback for tolerance management, human-in-the-loop and augmented-reality–enabled interaction, and vision–language–based constraint-aware reasoning with simulation- and reinforcement-learning-based task-and-motion planning. Rather than relying on idealized assumptions, the system explicitly addresses construction-relevant challenges such as geometric variability, tight tolerances, safety constraints, limited access, and coordination between multiple robots and human operators. Validation progresses from simulation and laboratory-scale experimentation to full-scale pilot demonstrations at realistic construction scales, establishing the CMRG as a unique testbed for exploring advanced construction automation workflows. From an industrial perspective, the project provides a critical bridge between autonomy research and construction-scale validation. The Korean SME partner will be able to develop and test autonomy algorithms, planning frameworks, and large-scale construction applications on a platform that supports realistic geometry, payload, and cooperation demands. By leveraging the research institute’s expertise in construction engineering, physical system integration, closed-loop execution, and experimental validation, the project will generate pilot-validated workflows, performance benchmarks, and system integration guidelines. These outcomes will reduce technical and engineering risk for future adoption and commercialization, strengthen the competitiveness of Korean construction-technology and robotics SMEs, and establish a foundation for next-generation, scalable automation in modular and prefabricated construction.

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6. References

6.1 Examples of Prior Cooperative Robotics Construction Projects

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6.3 Examples of Sensor- and AI-Based Weld Inspection Projects

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